Haodong Liang

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Education

University of California, Davis, USA. Ph.D. Electrical and Computer Engineering — Sep 2023 – Present

- Advisors: Lifeng Lai, Krishnakumar Balasubramanian
- GPA: 4.0/4.0

University of Michigan, Ann Arbor, USA. MSc. Statistics — Sep 2021 – May 2023

- GPA: 3.95/4.0

Xiamen University, China. BSc. Mathematical Statistics — Sep 2016 – Jun 2020

- GPA: 3.64/4.0. Major GPA: 3.8/4.0.

Research Interests

My research focuses on the theoretical and mechanistic foundations of in-context learning, a key capability of large language models (LLMs). I study how Transformers implement machine learning and decision-making algorithms in a provable view. My broader interests include continual learning, knowledge retention during fine-tuning, and differential privacy.

Research Projects

Provable In-Context Learning Capabilities of Transformer Models — Jan 2024 - Present, UC Davis

- Studied the theoretical and mechanistic foundations of how a pretrained Transformer model learns and executes algorithms in context, supported by empirical validations.
- Established that Transformers can handle endogeneity in in-context linear regression using instrumental variables, with provable excess loss bounds after pretraining.
- Established that a Transformer agent can improve its policy via SARSA or actor-critic in the in-context reinforcement learning setting, with provable convergence guarantees to the optimal parameter manifold.

Style Learning Induces Semantic Forgetting in Fine-Tuning — Mar 2026 - Present, UC Davis

- Investigated why supervised fine-tuning (SFT) can cause more semantic forgetting than reinforcement fine-tuning (RFT), even when teacher demonstrations are semantically correct.
- Derived an exact decomposition of semantic and style updates for SFT and RFT in a linear-softmax classification model, identifying teacher-induced style drift as a mechanism for semantic forgetting.
- Proved a finite-time lower bound on the forgetting gap between SFT and RFT under population updates, supported by empirical simulations on sequential classification tasks.

Differential Privacy under Endogeneity — Mar 2025 - Sep 2025, UC Davis

- Proposed a noisy two-stage gradient descent algorithm that addresses endogeneity while guaranteeing ρ-zero-concentrated differential privacy.
- Established finite-sample convergence rates and error bounds that quantify the trade-offs among optimization, privacy, and sampling error.
- Demonstrated practical privacy-utility trade-offs on both synthetic and real-world datasets.

Selected Publications

Learning Style, Forgetting Semantics: A Case Study of SFT and RFT on Classification Tasks 2026
H. Liang, Y. Jin, K. Balasubramanian, L. Lai
arXiv preprint, arXiv:2610.02437 | Paper

Transformers Provably Implement In-Context Reinforcement Learning with Policy Improvement 2026
H. Liang, L. Lai
Advances in Neural Information Processing Systems (NeurIPS 2026) | Paper

Differentially Private Two-Stage Gradient Descent for Instrumental Variable Regression 2025
H. Liang, Y. Jin, K. Balasubramanian, L. Lai
International Conference on Learning Representations (ICLR 2026) | Paper

Transformers Handle Endogeneity in In-Context Linear Regression 2024
H. Liang, K. Balasubramanian, L. Lai
International Conference on Learning Representations (ICLR 2025) | Paper

ASGARD is A Single-cell Guided Pipeline to Aid Repurposing of Drugs 2023
B. He, Y. Xiao, ***H. Liang***, Q. Huang, Y. Du, Y. Li, D. Garmire, D. Sun, L. Garmire
Nature Communications 14, 993 (2023) | Paper

Professional Services

Reviewer: ICLR (2026, 2027), NeurIPS (2026), IEEE Transactions on Information Theory.
Teaching Assistant: EEC 161 (Applied Probability), EEC 018 (Digital System Design), EEC 266 (Information Theory and Coding).
Consultant: CSCAR (Consulting for Statistics, Computing & Analytics Research).
Mentor: Maxwell Lin (Princeton University), Kaleb Kokott (UC Davis).

Honors and Awards

Shu Lin Graduate Student Conference Travel Award UC Davis, 2026
Summer Graduate Student Researcher Award UC Davis, 2025
University Graduate Program Fellowship UC Davis, 2023

Work Experience

BaishanCloud Technology Jul 2020 - Jul 2021
Data Analyst - Full time Xiamen, Fujian, China

- Designed optimization models and operational metrics for network bandwidth allocation, minimizing infrastructure cost while satisfying service quality constraints.
- Developed Python-based data pipelines integrated with internal APIs to automatically collect, process, and monitor network utilization metrics on a daily basis.
- Achieved 14% cost reduction (∼$1.2M annual savings).

Skills

Programming Languages: Python, R, Matlab, C++, Julia.
Frameworks & Libraries: PyTorch, Hugging Face Transformers, NumPy, SciPy.
Tools: Git, Linux, LaTeX.
Languages: English, Mandarin.